03/01/2024

Code-A_(Phase-1)

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MM : 720

NCERT Booster Test Series-2023-24_Test-03A

Time : 200 Min.

Topics Covered:
Physics : Semiconductor, Units and Measurements, Motion in a straight line, Motion in a Plane
Chemistry : Aldehydes, Ketones and Carboxylic Acids, Amines, Biomolecules, Some basic concepts of chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties
Botany : Cell: The Unit of Life, Cell Cycle and Cell Division
Zoology : Reproductive Health, Human Reproduction, Structural Organisation in Animals: Animal Tissues only, Biomolecules-I (Upto Nucleic acids included)

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B.** You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. The value of 5.3435942... up to 4 significant figures is

(1) 5.344
(2) 5.343
(3) 5.342
(4) 5.345

2. Find the relative error in X, if $X = \frac{A^3B^{2.3}}{C^2D^{3/2}}$

(1) $\frac{1}{3}\left(\frac{\Delta A}{A}\right) + \frac{2}{3}\left(\frac{\Delta B}{B}\right) + \frac{1}{2}\left(\frac{\Delta C}{C}\right) + \frac{3}{2}\left(\frac{\Delta D}{D}\right)$

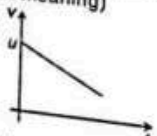
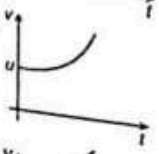
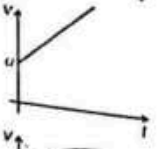
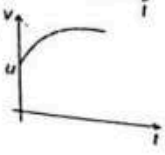
(2) $3\left(\frac{\Delta A}{A}\right) + \frac{2}{3}\left(\frac{\Delta B}{B}\right) + 2\left(\frac{\Delta C}{C}\right) + \frac{3}{2}\left(\frac{\Delta D}{D}\right)$

(3) $3\left(\frac{\Delta A}{A}\right) + \frac{2}{3}\left(\frac{\Delta B}{B}\right) - 2\left(\frac{\Delta C}{C}\right) - \frac{3}{2}\left(\frac{\Delta D}{D}\right)$

(4) $3\left(\frac{\Delta A}{A}\right) + \frac{3}{2}\left(\frac{\Delta B}{B}\right) + 2\left(\frac{\Delta C}{C}\right) + \frac{3}{2}\left(\frac{\Delta D}{D}\right)$

3. The dimensional formula for coefficient of viscosity is
(1) $[ML^{-1}T^{-2}]$
(2) $[M^{-1}L^{-1}T^{-1}]$
(3) $[ML^{-1}T^{-1}]$
(4) $[M^{2}L^{-1}T^{-1}]$

4. A car is moving with speed of 25 m/s. At time t = 5s, it applied brakes and comes to rest in next 5s. The average acceleration of the car from t = 5s to t = 10s is
(1) 5 m/s^2
(2) – 5 m/s^2
(3) – 4 m/s^2
(4) 4 m/s^2

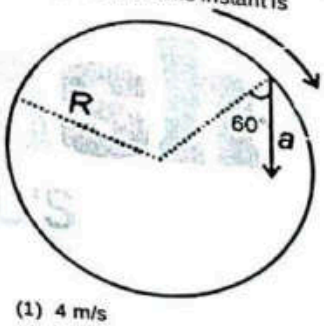
5. The relation $v = u + at$ is graphically represented as (where symbols have their usual meaning)
(1) 
(2) 
(3) 
(4) 

6. Two parallel rail tracks run north south. Train A moves towards north with a speed of 36 km/h and train B moves towards south with a speed of 72 km/h. Which of the following statements is correct?
(1) Velocity of B w.r.t. A is 30 m/s towards south
(2) Velocity of ground w.r.t. B is 20 m/s towards north
(3) Velocity of ground w.r.t. B is 20 m/s towards south
(4) Both (1) and (2)

7. If $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} - \hat{k}$, then a unit vector in the direction of $\vec{A} + \vec{B}$ is
(1) $\frac{4\hat{i}+7\hat{j}-3\hat{k}}{\sqrt{74}}$
(2) $\frac{2\hat{i}+3\hat{j}-\hat{k}}{\sqrt{14}}$
(3) $\frac{4\hat{i}-7\hat{j}-3\hat{k}}{\sqrt{74}}$
(4) $\frac{4\hat{i}+7\hat{j}+3\hat{k}}{\sqrt{74}}$

8. If $\vec{A} = 4\hat{i} + 4\hat{j}$, After rotating $\vec{A}$ by 45° anticlockwise then new vector obtained is
(1) $4\hat{i}$
(2) $4\hat{j}$
(3) $4\sqrt{2}\hat{i}$
(4) $4\sqrt{2}\hat{j}$

9. In the given figure, a = 20 m/s^2 represents the total acceleration of a particle moving in the clockwise direction in a circle of radius R = 3.6 m at a given instant of time. The speed of the particle at this instant is



(1) 4 m/s
(2) 5 m/s
(3) 6 m/s
(4) 7 m/s

10. At what angle from horizontal, maximum height and horizontal range are equal for ground to ground projectile motion
(1) $\theta = \tan^{-1}(4)$
(2) $\theta = \cot^{-1}(4)$
(3) $\theta = \tan^{-1}(6)$
(4) $\theta = \tan^{-1}(2)$

11. A particle follows a parabolic trajectory given by $y = \frac{x}{\sqrt{3}} - \frac{x^2}{6}$. Initial speed of particle is (Take g = 10 m/s^2)

(1) $4\sqrt{10}$ m/s

(2) $2\sqrt{10}$ m/s

(3) $3\sqrt{10}$ m/s

(4) $\sqrt{10}$ m/s

12. The position of a particle is given by $\vec{r} = 4t\,\hat{i} + 5t^2\,\hat{j} + 6t\,\hat{k}$ Magnitude of acceleration of particle at t = 1 s is

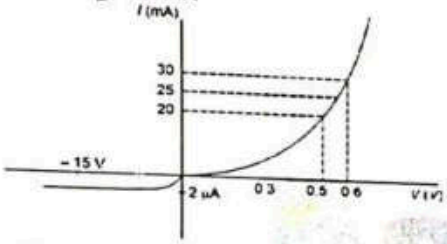
(1) 15 m/s^2

(2) $\sqrt{152}$ m/s^2

(3) 10 m/s^2

(4) 5 m/s^2

13. The *V-I* characteristic of a silicon diode is shown in the figure. What is the resistance of diode at I_D = 25 mA



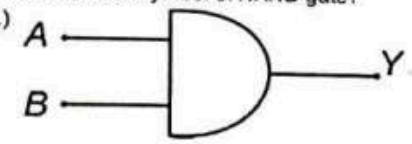
(1) 20 Ω

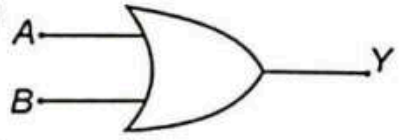
(2) 25 Ω

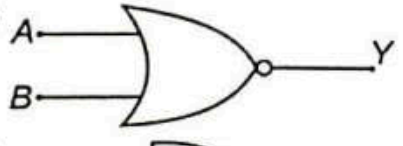
(3) 10 Ω

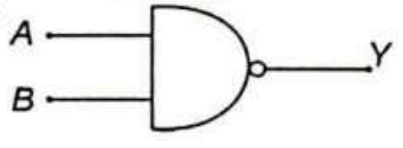
(4) 30 Ω

14. Which is correct symbol of NAND gate?

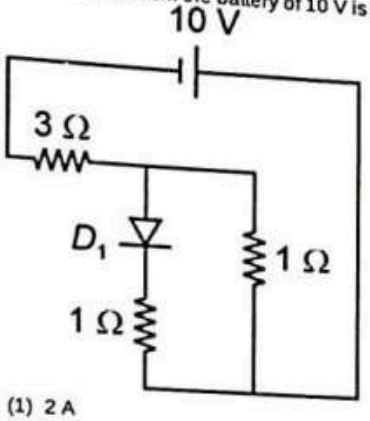
(1)


(2)


(3)


(4)


15. An arrangement of an ideal diode and resistances are shown to make a circuit. The current drawn from the battery of 10 V is



(1) 2 A

(2) $\frac{20}{7}$ A

(3) 2.5 A

(4) 3.3 A

16. An intrinsic semiconductor is converted into P-type extrinsic semiconductor by doping it with

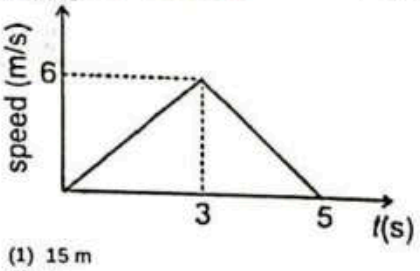
(1) Phosphorus

(2) Aluminium

(3) Arsenic

(4) Antimony

17. The speed-time graph of a particle moving along a curve is as shown below. The distance travelled by the particle from t = 0 to t = 5 s is



(1) 15 m

(2) 9 m

(3) 6 m

(4) 30 m

18. If force F, area A and density D are taken as the fundamental quantities, then mass can be represented as

(1) $[F^0A^{-1}D]$

(2) $[F^0AD^{-1}]$

(3) $[F^0AD]$

(4) $[F^0A^{3/2}D]$

19. 4 second (unit of angle) is equivalent to

(1) $\left[\frac{1}{900}\right]^{\circ}$

(2) $\left(\frac{1}{300}\right)^{\circ}$

(3) $\left[\frac{1}{200}\right]^{\circ}$

(4) $\left[\frac{1}{15}\right]^{\circ}$

20. A force of 40 N acts on a body. If the units of mass and length are doubled and the unit of the time tripled, then the force in new system becomes

(1) 90 N

(2) 90 new unit

(3) $\frac{160}{9}$ new unit

(4) $\frac{160}{9}$ N

21. If absolute error in a quantity A is ΔA and in quantity B is ΔB, then maximum possible error in calculation of $(A - B)$ will be

(1) $\Delta A - \Delta B$

(2) $\Delta B - \Delta A$

(3) $\frac{\Delta A}{A} + \frac{\Delta B}{B}$

(4) $\Delta A + \Delta B$

22. In equation $y = A\cos^2(\omega t - kx)$, where t is time and x is distance, dimensional formula of ω/k should be

(1) $[LT^{-1}]$

(2) $[L^{-1}T]$

(3) $[L^1T^1]$

(4) $[L^0T^0]$

23. A car travels a distance A to B at speed of 40 km h^{-1} and return to A at a speed of 30 km h^{-1}. What is the average velocity for whole journey?

(1) Zero

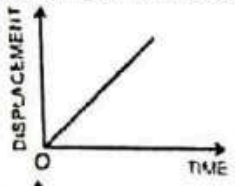
(2) 34.3 km h^{-1}

(3) 68.6 km h^{-1}

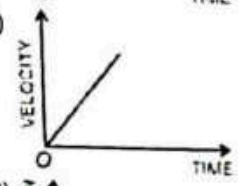
(4) 20 km h^{-1}

24. Which of the following graphs represents an object having uniform motion?

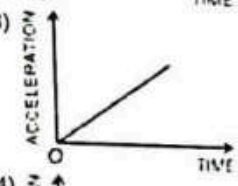
(1)



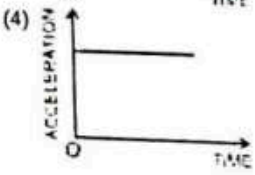
(2)



(3)



(4)



25. The position of a particle along x-axis at time t is given by $x = 1 + t - t^2$. The distance travelled by the particle in first 2 seconds is

(1) 1 m

(2) 2 m

(3) 2.5 m

(4) 3 m

26. The velocity of a bullet is reduced from 400 ms^{-1} to 100 ms^{-1} while travelling through material of thickness of 1 m. The average retardation will be

(1) 7.5×10^4 m/s^2

(2) 8.5×10^3 m/s^2

(3) 7.5×10^2 m/s^2

(4) 8.5×10^2 m/s^2

27. Two identical balls A and B are thrown from the top of a tower. Ball A is thrown upwards with speed v and ball B is thrown downwards with speed v, then

(1) Speed of ball A will be more than ball B at the ground

(2) Speed of ball B will be more than ball A at the ground

(3) Both balls A and B strike the ground with same speed

(4) Speed of ball A may be greater or less than speed of ball B at the ground

28. The range of a projectile at an angle of projection 37° from horizontal is R. For the same speed, if angle of projection is 45°, from horizontal then range becomes

(1) $\frac{25}{24}R$

(2) R

(3) $\frac{24}{25}R$

(4) $\frac{R}{4}$

29. A man crosses a river of width 400 m flowing at 40 m/s in minimum time. If velocity of man in still water is 50 m/s, then the displacement of man while he reaches the opposite bank of the river is.

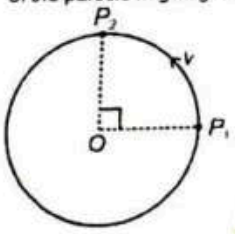
(1) 80 m

(2) $8\sqrt{4}\,\text{m}$

(3) $80\sqrt{41}\,\text{m}$

(4) $10\sqrt{41}\,\text{m}$

30. A particle of mass m is moving with a uniform speed v on a circular path of radius R. The magnitude of change in acceleration of the particle in going from P_1 to P_2 is



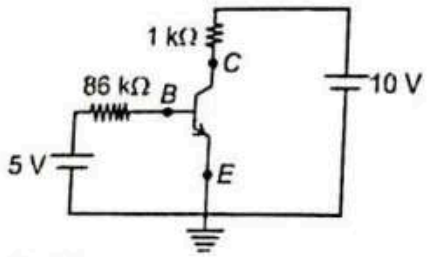
(1) Zero

(2) $\frac{v^3}{2R}$

(3) $\frac{2v^2}{R}$

(4) $\frac{\sqrt{2}\,v^2}{R}$

31. A CE amplifier circuit is shown in the figure. If $\beta = 100$ and $V_{BE} = 0.7$ V, then value of V_{CE} will be



(1) 7 V

(2) 8 V

(3) 3 V

(4) 5 V

32. Boolean expression $Y = \overline{\overline{A} + \overline{B}}$, can also be written as

(1) $A \cdot B$

(2) $\overline{A} \cdot B$

(3) $\overline{A} + \overline{B}$

(4) $\overline{A \cdot B}$

33. In a full wave rectifier circuit operating from 50 Hz mains frequency, the frequency in the ripple would be

(1) 50 Hz

(2) 25 Hz

(3) 100 Hz

(4) 707 Hz

34. Consider the following statements.

(a) P-type semiconductors are positively charged.

(b) N-type semiconductors are negatively charged.

(c) In N-type semiconductor $n_e > n_h$.

(d) In P-type semiconductor $n_e > n_h$.

Choose the incorrect statements.

(1) (c) only

(2) (a), (b) and (d) only

(3) (a), (c) and (d) only

(4) All

35. For transistor action (amplification), which of the following statement is correct?

(1) Base emitter and collector regions should have similar size and doping concentration

(2) Base region must be very thin and lightly doped

(3) Both junctions should be forward biased

(4) Both junctions should be reversed biased

SECTION-B

36. An insect trapped in a circular groove of radius 20 cm moves along the grove steadily and completes 14 revolutions in 44 s then linear speed of the motion is

(1) 0.2 m/s

(2) 0.3 m/s

(3) 0.4 m/s

(4) 0.5 m/s

37. Rain is falling vertically with a speed of 20 m/s. A man rides a bicycle with a speed of 10 m/s in the north to south direction. Then velocity of rain w.r.t. man is

(1) $20\sqrt{5}$ m/s

(2) $10\sqrt{5}$ m/s

(3) $5\sqrt{10}$ m/s

(4) $2\sqrt{5}$ m/s

38. Suppose a pure Silicon (Si) crystal has 6×10^{27} atoms m^{-3}. It is doped by 1 ppm concentration of pentavalent impurity. Calculate number of holes [Given that $n_i = 1.2 \times 10^{15}\ m^{-3}$.]

(1) $2.4 \times 10^{9}\ m^{-3}$

(2) $2.4 \times 10^{8}\ m^{-3}$

(3) $4.8 \times 10^{8}\ m^{-3}$

(4) $4.8 \times 10^{9}\ m^{-3}$

39. Which of the following P-n junction diodes is forward-bias?

(1) 2 V — 3 V

(2) − 5 V — − 2 V

(3) 0 V — 3 V

(4) − 2 V — − 5 V

40. The percentage error in the measurement of g is (given that $g = \frac{4\pi^2 L}{T^2}$ $L = (20 \pm 0.2)$ cm, $T = (200 \pm 1)$ s)

(1) 1%

(2) 2%

(3) 3%

(4) 4%

41. A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 1 mm

Circular scale reading : 48 divisions

Given that 2 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is

(1) 1.96 mm

(2) 1.96 cm

(3) 0.96 mm

(4) 0.96 cm

42. The dimension of $\frac{B^2}{2\mu_0}$, where μ_0 is magnetic permeability of free space and B is magnetic field, is

(1) $[ML^2T^{-2}]$

(2) $[ML^2T^{-1}]$

(3) $[ML^{-1}T^{-2}]$

(4) $[MLT^{-1}]$

43. A ball is thrown vertically downward with a velocity of 30 m/s from the top of a tower. It hits the ground after some time with a velocity of 70 m/s. The height of the tower is : ($g = 10\ m/s^2$)

(1) 100 m

(2) 200 m

(3) 120 m

(4) 220 m

44. The equation of state of gas (hypothetical system) is given by, $\left(P + \frac{\alpha}{V^2}\right) = Q\frac{\theta^2}{V}$ where P is the pressure, V is volume, θ is absolute temperature and Q and α are dimensional constant. The dimensional formula of $\left(\frac{\alpha}{Q}\right)$ is

(1) $[LT^3K^2]$

(2) $[L^3K]$

(3) $[ML^3K]$

(4) $[L^3K^2]$

45. Position of particle A varies as $x_A = 2t^2 - 4t$ and that of B as $x_B = t^3 - 2t$. Find velocity of B with respect to A at $t = 4$ s.

(1) 36 m/s

(2) 34 m/s

(3) 58 m/s

(4) 38 m/s

46. A particle moving with uniform acceleration covers 25 m in 5th second and 35 m in 7th second of its journey, then distance travelled in its 3rd second will be

(1) 15 m

(2) 20 m

(3) 30 m

(4) 10 m

47. A boy travels d distance towards north with speed v_1 then takes a right turn towards east to travel same distance d with speed v_2. Find average velocity of entire journey.

(1) $\frac{\sqrt{2}v_1v_2}{v_1+v_2}$

(2) $\frac{2v_1v_2}{v_1+v_2}$

(3) $\sqrt{v_1v_2}$

(4) $\frac{v_1+v_2}{2}$

48. A particle is moving on a circle of radius A. At an instant its speed and tangential acceleration are B and C respectively. Total acceleration of the particle at that instant is

(1) $\sqrt{\frac{A^2}{B^2}+C^2}$

(2) $\sqrt{\frac{A^4}{B^2}+C^2}$

(3) $\sqrt{\frac{B^4}{A^2}+C^2}$

(4) $\frac{B^2}{A}+C$

49. A potential difference of 2 V is applied between the opposite faces of a Ge crystal plate of area 1 cm^2 and thickness 0.5 mm. If the concentration of electrons in Ge is $2 \times 10^{19}/m^3$ and mobilities of electrons and holes are 0.36 $m^2V^{-1}s^{-1}$ and 0.14 $m^2V^{-1}s^{-1}$ respectively, then the current flowing through the plate will be

(1) 0.25 A

(2) 0.45 A

(3) 0.56 A

(4) 0.64 A

50. In a semiconductor $\frac{5}{6}$ of the total current is due to flow of electrons. If drift velocity of electrons is $\frac{2}{3}$ times that of holes, then the ratio of the concentration of holes and electrons is

(1) 15 : 2

(2) 2 : 3

(3) 2 : 15

(4) 3 : 2

CHEMISTRY

SECTION-A

51. The frequency of electromagnetic radiation of wave number 10^4 cm^{-1} is

(1) 2×10^{13} Hz

(2) 3×10^{14} Hz

(3) 3×10^{12} Hz

(4) 2×10^{12} Hz

52. A photon of wavelength 400 nm is absorbed by a gas and then emitted as two photons. One photon has a wavelength 800 nm. The wavelength of other photon will be

(1) 700 nm

(2) 600 nm

(3) 800 nm

(4) 900 nm

53. Consider the following statements for photoelectric effect.

(a) The number of electrons ejected is proportional to the intensity or brightness of light.

(b) There is a time lag between the striking of light beam and the ejection of electrons from the metal surface.

(c) For each metal there is a characteristic maximum frequency known as threshold frequency above which photoelectric effect is not observed.

The incorrect statement(s) is/are

(1) (a) only

(2) (a) & (b) only

(3) (b) & (c) only

(4) (a), (b) & (c)

54. The radius ratio of 3rd orbit of He^+ to 2nd orbit of Li^{2+} is
(1) $\left(\frac{3}{2}\right)^3$
(2) $\left(\frac{2}{3}\right)^2$
(3) $\left(\frac{1}{2}\right)^2$
(4) $\left(\frac{3}{2}\right)^2$

55. The amount of energy absorbed in the transition n = 1 to n = 2 in Li^{2+} ion will be
(1) 33.9 eV
(2) 28.7 eV
(3) 54.6 eV
(4) 91.8 eV

56. Match List-I with List-II and choose the correct code.

	List-I (Orbitals)		List-II (Number of radial nodes)
(a)	2p	(i)	1
(b)	3p	(ii)	3
(c)	3s	(iii)	0
(d)	4s	(iv)	2

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

57. The electronic configuration of an element (X) is $1s^2, 2s^2\ 2p^6, 3s^1$. The atomic number of the element which is just below X in the periodic table is
(1) 19
(2) 20
(3) 18
(4) 12

58. If the name of the element according to IUPAC nomenclature is Unniloctium then its atomic number and symbol will be
(1) 105, Uup
(2) 112, Uno
(3) 108, Uno
(4) 109, Une

59. The correct decreasing order of size will be
(1) $K^+ > Ca^{2+} > S^{2-} > Cl^-$
(2) $K^+ > Ca^{2+} > Cl^- > S^{2-}$
(3) $S^{2-} > Cl^- > K^+ > Ca^{2+}$
(4) $S^{2-} > Ca^{2+} > K^+ > Cl^-$

Question type: Paragraph Based from Q60 to Q61
An **empirical formula** represents the simplest whole number ratio of various atoms present in a compound, whereas, the **molecular formula** shows the exact number of different types of atoms present in a molecule of a compound.
If the mass per cent of various elements present in a compound is known, its empirical formula can be determined. Molecular formula can further be obtained if the molar mass is known.

60. An organic compound contains 40% carbon, 53.33% oxygen and rest hydrogen. The molecular formula of the compound can be
(1) C_2H_4O
(2) $C_2H_4O_2$
(3) $C_2H_6O_2$
(4) C_2H_6O

61. For which of the following compound empirical formula and molecular formula are same?
(1) CH_3COOH
(2) $HOCH_2CH_2OH$
(3) C_6H_6
(4) CH_3CHO

62. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): 0.031 has four significant figures.
Reason (R): Zeros preceding to first non-zero digit are significant.
In the light of above statements choose the correct answer.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

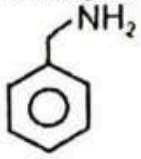
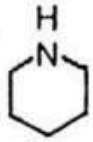
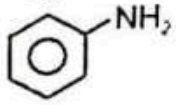
63. If 12 g of H_2 reacts with 28 g of N_2 to give ammonia. Then
(1) H_2 is the limiting reagent
(2) 6 g of H_2 is left
(3) 8 g of H_2 is left
(4) 17 g of NH_3 is formed

64. Match List-I with List-II and choose the correct code.

	List-I (Vitamins)		List-II (Deficiency diseases)
(a)	Pyridoxin	(i)	Cheilosis
(b)	Riboflavin	(ii)	Muscular weakness
(c)	Ascorbic acid	(iii)	Convulsions
(d)	Vitamin E	(iv)	Scurvy

(1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

65. Which among the following will not react with Hinsberg's reagent?

(1) 
(2) 
(3) 
(4) 

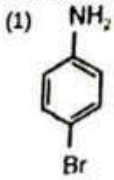
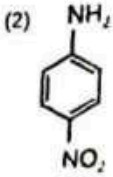
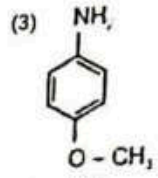
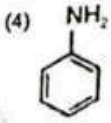
66. The amine which is most basic in aqueous solution is
(1) $(CH_3)_2NH$
(2) $(CH_3)_3N$
(3) CH_3NH_2
(4) NH_3

67. Consider the reaction sequence

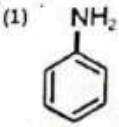
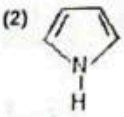
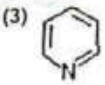
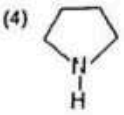
$$CH_3 - CN \xrightarrow{H_3O^+} A \xrightarrow[\Delta]{NH_3} B \xrightarrow{Br_2 / KOH} C$$

C is
(1) Ethanamine
(2) Methanamine
(3) Methyl bromide
(4) Ethyl bromide

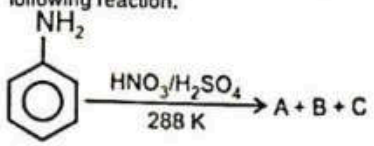
68. The arylamine which undergoes diazotization most readily is

(1) 
(2) 
(3) 
(4) 

69. Which among the following will give white precipitate with $Br_2 + H_2O$?

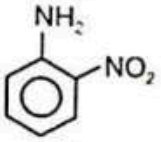
(1) 
(2) 
(3) 
(4) 

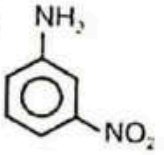
70. Find out the product with least yield in the following reaction.



(1) All have equal yield

(2) 

(3) 

(4) 

71. Which of the following oxides is neutral?

(1) CaO

(2) N_2O

(3) SO_3

(4) Cl_2O_7

72. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Fluorine has less negative electron affinity than chlorine.

Reason (R): There is relatively greater effectiveness of 2p electrons in the small F atom to repel the additional electron entering the atom than done by 3p electrons in the larger Cl atom.

In the light of above statements choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

73. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Diagonal relationship is shown between Be and Al.

Reason (R): First ionization energy of Be is equal to that of Al.

In the light of above statements choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

74. According to modern periodic law "The physical and chemical properties of elements are periodic functions of their _____"

(1) Atomic number

(2) Mass number

(3) Atomic mass

(4) Number of neutrons

75. Consider the following reaction sequence

$$CH_3CONH_2 \xrightarrow[\Delta]{P_2O_5} A \xrightarrow{H_3O^+} B$$

Product B is

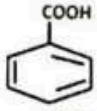
(1) Acetaldehyde

(2) Acetic acid

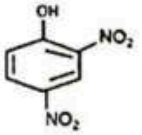
(3) Ethylamine

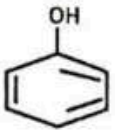
(4) Ethyl alcohol

76. The compound which will not react with $NaHCO_3$ is _____

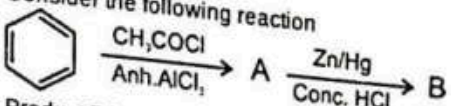
(1) 

(2) 

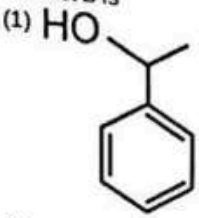
(3) 

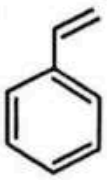
(4) 

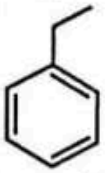
77. Consider the following reaction

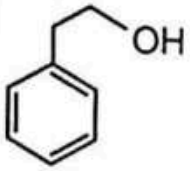


Product B is

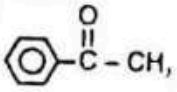
(1) 

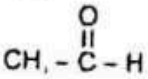
(2) 

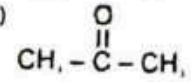
(3) 

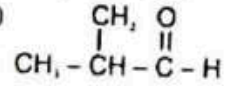
(4) 

78. Which of the following will not give aldol condensation?

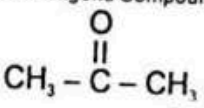
(1) 

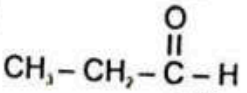
(2) 

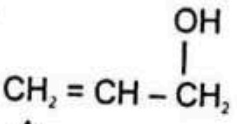
(3) 

(4) 

79. An organic compound (A) with molecular formula C_3H_6O gives an orange red precipitate with 2,4-DNP and it also reduces Tollens' reagent. Compound A is

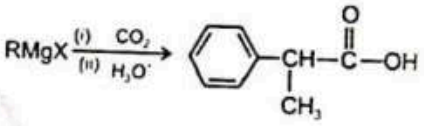
(1) 

(2) 

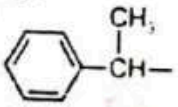
(3) 

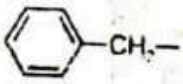
(4) 

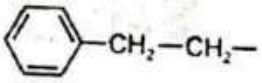
80. Consider the reaction

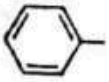


Identify R

(1) 

(2) 

(3) 

(4) 

81. When ethylbenzene is reacted with alkaline or acidic $KMnO_4$ major product formed is

(1) 2-phenylethanol
(2) 2-phenylethanoic acid
(3) Benzoic acid
(4) Benzaldehyde

82. The compound which will not react with Tollens' reagent is

(1) D-Mannose
(2) D-Fructose
(3) Lactose
(4) Sucrose

83. Select the correct statement in the following
(1) Amylose and amylopectin are components of starch
(2) Amylose is water insoluble
(3) Amylopectin is water soluble
(4) Amylose is branch chain polymer of β-D-glucose

84. A non-essential amino acid among the following is
(1) Valine
(2) Leucine
(3) Lysine
(4) Glycine

85. Mannose and glucose are
(1) Epimers
(2) Functional Isomers
(3) Tautomers
(4) Anomers

SECTION-B

86. Given below are two statements:
Statement I: There are five *d*-orbitals designated as d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$ and d_{z^2}. The shapes of first four *d*-orbitals are similar to each other, whereas that of the fifth one, d_{z^2}, is different from others.
Statement II: The *d* orbitals for which n is greater than 3 (4*d*, 5*d*...) also have shapes similar to 3*d* orbital, but differ in energy and size.
In the light of above statements choose the correct answer.
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

87. The correct order of energies of 2s orbital for given alkali metals is
(1) $E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$
(2) $E_{2s}(K) > E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na)$
(3) $E_{2s}(H) > E_{2s}(K) > E_{2s}(Na) > E_{2s}(Li)$
(4) $E_{2s}(K) > E_{2s}(Na) > E_{2s}(Li) > E_{2s}(H)$

88. Assume the first four ionization enthalpy values of an element are 284, 412, 656 and 3210 kJ mol^{-1}. The number of valence electrons in the element is
(1) 1
(2) 2
(3) 3
(4) 4

89. The correct order of second ionization energy for Li, B, Be and C is
(1) Li > B > C > Be
(2) Li > C > B > Be
(3) Be > B > C > Li
(4) Be > Li > C > B

90. The group number and period number respectively of an element with atomic number 72 is
(1) 3, 5
(2) 4, 6
(3) 4, 7
(4) 3, 6

91. 200 mL of 0.1 M H_2SO_4 and 300 mL of 0.2 M HCl are mixed together. The molarity of H^+ in the resultant mixture is
(1) 0.2 M
(2) 0.1 M
(3) 0.02 M
(4) 0.01 M

92. 400 g sample of limestone liberates 44 g carbon dioxide on heating. The percentage purity of $CaCO_3$ in the sample of limestone is
(1) 25%
(2) 30%
(3) 50%
(4) 75%

93. 32 g of methyl alcohol is present in 132 g of an aqueous solution. The molality of methyl alcohol in the solution is

(1) 5 m

(2) 8 m

(3) 12 m

(4) 10 m

94. The maximum number of electrons present in the orbitals having quantum numbers n = 3 and l = 1 are

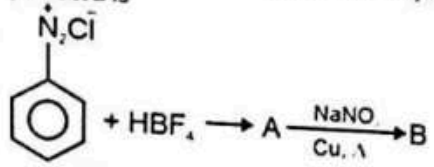
(1) 2

(2) 4

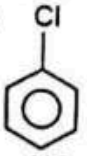
(3) 10

(4) 6

95. In the given reaction sequence, major product B is



(1) 

(2) 

(3) 

(4) 

96. Which of the following compounds on reaction with Grignard reagent gives primary alcohol?

(1) HCHO

(2) RCHO

(3) R_2CO

(4) RCOOH

97. Which among the following carboxylic acids has the highest pK_a value?

(1) HCOOH

(2) C_6H_5COOH

(3) O_2NCH_2COOH

(4) CH_3COOH

98. Number of 3° amines possible for $C_5H_{13}N$ is

(1) 1

(2) 2

(3) 3

(4) 5

99. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Glycine is the only amino acid which is optically inactive.

Reason (R): Glycine has no chiral carbon.

In the light of above statements choose the correct answer.

(1) Both assertion (A) and reason (R) are correct statements, and reason is the correct explanation of the assertion.

(2) Both assertion (A) and reason (R) are correct statements, but reason is not the correct explanation of the assertion.

(3) Assertion (A) is correct, but reason (R) is incorrect.

(4) Assertion (A) is incorrect, but reason (R) is correct.

100. The linkage between the two monosaccharide units in lactose is

(1) C_1 of β-D glucose and C_4 of β-D galactose

(2) C_1 of β-D galactose and C_4 of β-D glucose

(3) C_1 of α-D galactose and C_4 of β-D glucose

(4) C_1 of β-D galactose and C_4 of α-D glucose

BOTANY

SECTION-A

101. Select the **incorrect** statement about glycocalyx in bacteria.
(1) Can be present in the form of a loose sheath
(2) It is the outermost layer of cell envelope
(3) Differs in composition and thickness among different bacteria
(4) Is always hard and thick

102. What will be the amount of DNA in meiosis-II product, if the meiocyte contains 40 pg DNA in G_1 phase?
(1) 20 pg
(2) 80 pg
(3) 10 pg
(4) 5 pg

103. The middle lamella present between neighbouring cells of plants is mainly made up of
(1) Calcium carbonate
(2) Calcium pectate
(3) Hemicellulose
(4) Proteins

104. Nucleolus and nuclear membrane reappear during
(1) Anaphase
(2) Telophase
(3) Late prophase
(4) Metaphase

105. Organelle in eukaryotic cell that appears like a cartwheel
(1) Has peripheral doublets
(2) Is surrounded by membrane
(3) Is surrounded by amorphous pericentriolar materials
(4) Occurs in pair lying parallel to each other

106. Which of the following statements is **incorrect** w.r.t cytoskeleton?
(1) Helps in motility.
(2) Maintains the shape of cell.
(3) Is helpful in production of ATP.
(4) Provides mechanical support to the cell.

107. Select **wrong** statement with respect to significance of mitosis
(1) Maintains chromosome number
(2) Responsible for the growth of individual
(3) Introduces variations
(4) Play important role in healing and repair

108. In quiescent phase, cells
(1) Undergo rapid proliferation
(2) Are metabolically active
(3) Are in G_1 phase
(4) Continuously increase in size

109. If a cell has 48 chromosomes in G_1 phase then what will be the number of bivalents in prophase I ?
(1) 48
(2) 96
(3) 24
(4) 12

110. The plasma membrane consists mainly of
(1) Phospholipids embedded in protein bilayer
(2) Proteins and phospholipids
(3) Proteins and carbohydrates only
(4) Sugar embedded in glycolipids

111. Which phase lasts more than 95% of the duration of cell cycle?
(1) Telophase
(2) Diplotene
(3) Anaphase
(4) Interphase

112. Select the **mis-matched** pair.
(1) Vacuole – Tonoplast
(2) Inclusion body – Found only in eukaryotic cell
(3) Chromatophores – Blue-green algae
(4) Mitochondria – Sausage shaped

113. The function of ribosome of a polysome is to
(1) Translate the mRNA into protein
(2) Synthesise lipids in eukaryotic cells
(3) Replicate DNA during cell division
(4) Store various hydrolytic enzymes

114. A pair of synapsed homologous chromosomes is known as
(1) Bivalent
(2) Chiasmata
(3) Monad
(4) Monovalent

115. Zygotene is characterized by
(A) Pairing of homologous chromosomes
(B) Appearance of synaptonemal complex
(C) Formation of bivalents
(D) Formation of chiasmata
(1) (A), (B) & (D) only
(2) (B), (C) & (D) only
(3) All except (C)
(4) All except (D)

116. Which of the following scientist first saw and described a live cell?
(1) Robert Brown
(2) Rudolf Virchow
(3) Theodore Schwann
(4) A. V. Leeuwenhoek

117. Select the **incorrect** match.
(1) Robert Brown – Nucleus.
(2) Rudolf Virchow – Omnis cellula e cellula.
(3) Theodore Schwann – British Zoologist
(4) Matthias Schleiden – Studied different types of animal cell

118. Read the statements and select the **correct** option.
Statement A: Cytoplasm is the main arena of cellular activities in only plant cells.
Statement B: Ribosomes are non-membrane bound organelles.
(1) Only statement A is correct
(2) Both statements A and B are correct
(3) Only statement B is correct
(4) Both statements A and B are incorrect

119. Select the **correct** option w.r.t shape of cells.

A.	W.B.C.	(i).	Branched and long
B.	R.B.C.	(ii).	Round and biconcave
C.	Nerve cell	(iii).	Amoeboid
D.	Mesophyll cell	(iv).	Round and oval

(1) A-(iii), B-(ii), C-(i), D-(iv)
(2) A-(ii), B-(iii), C-(i), D-(iv)
(3) A-(iii), B-(ii), C-(iv), D-(i)
(4) A-(iv), B-(ii), C-(i), D-(iii)

120. Select the **incorrect** match w.r.t shape of bacteria.
(1) Bacillus – rod like
(2) Coccus – spherical
(3) Vibrio – comma shaped
(4) Spirillum – amoeboid

121. _____ confers certain unique phenotypic characters to bacteria
(1) Chromatophores
(2) Mesosomes
(3) Ribosomes
(4) Plasmid

122. Which of the following statements is **not correct** w.r.t. prokaryotic cell?
(1) Fimbriae are small bristle like fibres sprouting out of the cell which help in attachment to rocks.
(2) Flagellum is composed of three parts filament, hook and basal body.
(3) Mesosomes are formed by the extension of plasma membrane.
(4) Inclusion bodies are membrane bound structures present freely in the cytoplasm.

123. Read the following statements and select the **correct** option.
Statement A: Animal cells have centrioles which are absent in almost all plants cells.
Statement B: Plants cells possess cell walls, plastids and a large central vacuole which are absent in animal cells.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

124. Fluid mosaic model was proposed by
(1) Singer and Nicolson
(2) George Palade
(3) Camillo golgi
(4) Robertson

125. Select the **correct** option w.r.t set of organelles included in endomembrane system.
(1) E.R, golgi complex, lysosomes and vacuoles.
(2) E.R, golgi complex, lysosomes and Ribosomes.
(3) Lysosomes, Ribosomes, golgi complex, plastids.
(4) Mitochondria, Plastids, lysosomes and vacuoles.

126. Which of the following statements is **incorrect** w.r.t. Endoplasmic Reticulum.
(1) RER is actively involved in protein synthesis and secretion
(2) SER is the major site for synthesis of lipids
(3) It divides the intracellular space into three distinct compartments
(4) In animal cells lipid like steroidal hormones are synthesised in SER

127. The centromere is present at the terminal end of the chromosome in
(1) Metacentric chromosome
(2) Acrocentric chromosome
(3) Telocentric chromosome
(4) Sub-metacentric chromosome

128. Plastid which is specific for storage of oils and fats is
(1) Amyloplast
(2) Aleuroplast
(3) Elaioplast
(4) Chromoplast

129. Which of the following is **not** the function of mesosome in bacteria?
(1) DNA replication
(2) Cell wall formation
(3) Photosynthesis
(4) Respiration

130. Which of the following statement is **correct** for prophase
(1) Spindle fibres attach to kinetochores of chromosome
(2) Centromere split and chromatids separate
(3) Chromatids move to opposite poles
(4) Chromosomal material condenses to form compact chromosomes

131. Which of the following options **correctly** depicts the sequence of stages of prophase I of meiosis I.
(1) Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis.
(2) Leptotene → Pachytene → Diplotene → Zygotene → Diakinesis
(3) Diakinesis → Diplotene → Pachytene → Zygotene → Leptotene
(4) Pachytene → Diakinesis → Diplotene → Zygotene → Leptotene

132. The X-shaped structures formed by the homologous chromosomes in diplotene stage are
(1) Chiasmata
(2) Kinetochores
(3) Recombination nodule
(4) Telomeres

133. The interphase does **not** include
(1) G_1 phase
(2) G_2 phase
(3) M phase
(4) S phase

134. The mechanism by which conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms is
(1) Mitosis
(2) Meiosis
(3) Interphase
(4) Cytokinesis

135. Select the **incorrect** statement
(1) The beginning of diplotene is recognised by the dissolution of the synaptonemal complex
(2) Crossing over is the exchange of genetic material between two homologous chromosomes
(3) In oocytes of some vertebrates, diplotene can last for months or years
(4) Terminalisation of chiasmata occurs in leptotene

SECTION-B

136. Identify the statement that is **true** w.r.t. leptotene stage.
(1) Chromatin material begins to condense
(2) Duplication of DNA occurs
(3) Chromatids move towards opposite poles
(4) Recombination nodules are seen

137. Average duration of cell cycle of a typical human cell in culture is
(1) 90 minutes
(2) 24 hours
(3) 90 hours
(4) 24 minutes

138. The subunits of ribosomes present in prokaryotic cells are
(1) 50S and 30S
(2) 60S and 40S
(3) 50S and 40S
(4) 30S and 60S

139. Gas vacuoles
(a) Are found in all aquatic eukaryotes.
(b) Are membraneless.
(c) Provide buoyancy to the cell.
(1) Only (a) is true
(2) (a) and (b) are true
(3) (b) and (c) are true
(4) (a) and (c) are true

140. Final stage of nuclear division is
(1) Cytokinesis
(2) Metaphase
(3) Anaphase
(4) Telophase

141. Glycosidation of lipids, *i.e.*, formation of glycolipids occurs in
(1) Ribosomes
(2) Golgi apparatus
(3) Lysosome
(4) Chloroplasts

142. The arrangement of axonemal microtubules in eukaryotic cilia and flagella is
(1) 9 + 2
(2) 9 + 0
(3) 18 + 0
(4) 27 + 2

143. In human beings, the membrane of the RBCs has approximately
(1) 52% protein and 40% lipids
(2) 40% protein and 52% lipids
(3) 50% protein and 50% lipids
(4) 40% protein and 60% lipids

144. Which of the following cell organelle is double membrane bound?
(1) Lysosome
(2) Chloroplast
(3) Endoplasmic reticulum
(4) Ribosome

145. Most of the cell organelle's duplication occurs in
(1) G_2 phase
(2) G_1 phase
(3) S phase
(4) M phase

146. Na^+/K^+ pump in animal cells
(1) Requires energy
(2) Helps in passive transport
(3) Shows osmosis
(4) Allow movement of neutral solutes across the membrane along the concentration gradient

147. Prophase I is similar to prophase II as
(1) Both always occur in haploid cells
(2) Both include five sub stages
(3) Few cell organelles and nuclear membrane disappear in both
(4) Both are equally complex and prolonged stages

148. Disc shaped structure on the sides of centromere of each chromosome is called
(1) Secondary constriction
(2) Kinetochore
(3) Primary constriction
(4) Telomere

149. Synapsis occurs between
(1) Two sister chromatids
(2) Two homologous chromosomes
(3) Two non-homologous chromosomes
(4) Sex chromosomes only

150. Splitting of centromere is observed during
(1) Anaphase I
(2) Anaphase II
(3) Telophase I
(4) Metaphase I

ZOOLOGY

SECTION-A

151. Collagen fibres are secreted by A and these fibres are oriented differently in B connective tissue.
Choose the option that correctly identifies A and B respectively.
(1) Mast cells, cartilage
(2) Fibroblasts, dense regular
(3) Fibroblasts, dense irregular
(4) Macrophages, dense irregular

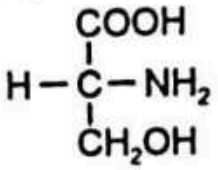
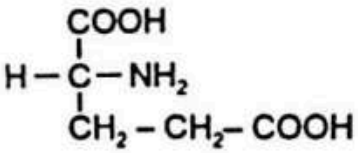
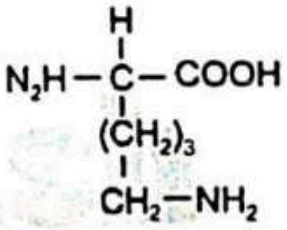
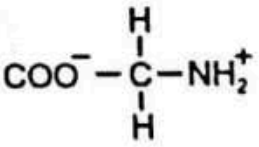
152. Select the **correct** statement w.r.t. neural tissue.
(1) All neuroglial cells are excitable in nature
(2) Neurons are unit of neural system that respond to stimulus
(3) Muscles constitute less than one-half the volume of neural tissue in our body
(4) In adult humans, neural tissue consists of actively dividing cells called neurons and glial cells

153. A segment of ds DNA has 100 adenine and 100 cytosine residues. The total number of nucleotides and hydrogen bonds present in it respectively are
(1) 400, 500
(2) 99, 500
(3) 400, 198
(4) 500, 400

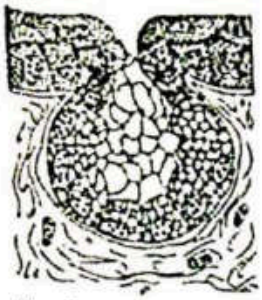
154. Read the following given statements w.r.t. cellulose and choose the **correct** option.
Statement A : It does not contain complex helices.
Statement B : It cannot hold I_2.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

155. How many nitrogen atoms are present in the structure of a substituted purine nitrogenous base which forms two hydrogen bonds with thymine?
(1) Six
(2) Four
(3) Five
(4) Three

156. Which of the following statements is **correct** w.r.t α-amino acids?
(1) A particular property of amino acids is the ionisable nature of $-NH_2$ and $-COOH$ groups.
(2) Tyrosine, phenylalanine and lysine are aromatic amino acids.
(3) They all are substituted ethane.
(4) Amino acids which occur in proteins are only of ten types

157. Consider the structures given below and select an acidic amino acid from the following.
(1)

(2)

(3)

(4)


158. The organic molecule 'A' is _____% of the total cellular mass in living organisms. It consists of monomeric units that are joined together with the help of peptide bonds.
Choose the option that fills the blank **correctly**.
(1) 5 – 7
(2) 10 – 15
(3) 3
(4) 2

159. 

Identify the type of gland shown in above figure and choose the option that cannot be considered as its example.

(1) Salivary gland
(2) Goblet cells
(3) Sweat gland
(4) Sebaceous gland

160. Choose the **correct** option which represents a set of the same type of tissues present in human.

(1) Neuron, tendon
(2) Bone, endothelium
(3) Tendon, ligament
(4) Areolar, muscular

161. Which of the following is the correct similarity between skeletal muscle fibres and cardiac muscle fibres?

	Parameter	Skeletal muscle fibres	Cardiac muscle fibres
(1)	Shape	Cylindrical	Cylindrical
(2)	Branching	Branched	Branched
(3)	Location of nuclei	Peripheral	Peripheral
(4)	Number of nuclei	Multi-nucleated	Multinucleated

(1) 1
(2) 2
(3) 3
(4) 4

162. Which of the following tissues is best to study all different types of specialized junctions that provide both structural and functional links between its individual cells?

(1) Epithelial tissue
(2) Skeletal muscle
(3) Chemical synapse in a neuron
(4) Blood

163. Match column I with column II w.r.t humans and choose the **correct** option.

	Column I		Column II
(a)	Cartilage	(i)	Attach skeletal muscles to bone
(b)	Bone	(ii)	Matrix is solid and pliable
(c)	Blood	(iii)	Calcium salts and collagen fibres
(d)	Tendons	(iv)	Main circulating fluid

(1) a(i) b(ii) c(iii) d(iv)
(2) a(ii) b(iii) c(iv) d(i)
(3) a(iii) b(ii) c(iv) d(i)
(4) a(ii) b(iii) c(i) d(iv)

164. In human females, tertiary follicle is characterised by the presence of a fluid filled cavity called

(1) Granulosa
(2) Antrum
(3) Theca interna
(4) Theca externa

165. In humans, the sperm motility is facilitated by energy releasing intracellular structure present in the _____ of sperms.
Select the option to fill the blank correctly.

(1) Neck
(2) Acrosome
(3) Middle piece
(4) Nucleus

166. Select the **correct** statement w.r.t. spermatogenesis in humans.

(1) Spermatogenesis is initiated during embryonic development stage.
(2) Androgens inhibit the process of spermatogenesis.
(3) Primary spermatocytes periodically undergo meiosis.
(4) Each spermatogonium is a haploid cell and contains 23 chromosomes.

167. Which of the following is incorrect w.r.t. 'Saheli'

(1) It is a 'once a week' pill
(2) It has high contraceptive value and very few side effects
(3) It was developed by scientists at CDRI, Lucknow
(4) It is a steroidal preparation

168. Choose the correct pathway of transport of spermatozoa in an adult human male.

(1) Seminiferous tubules → Vasa efferentia → Rete testis → Epididymis → Vas deferens → Ejaculatory duct → Urethra

(2) Seminiferous tubules → Rete testis → Vasa efferentia → Vas deferens → Epididymis → Ejaculatory duct → Urethra

(3) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra

(4) Seminiferous tubules → Rete testis → Vas deferens → Vasa efferentia → Epididymis → Ejaculatory duct → Urethra

169. In a human female, a cycle of 28 days exhibits three uterine phases. One of the phase is secretory phase and it exhibits a peak of hormone 'X' which helps in maintaining the endometrium. This hormone 'X' is secreted during pregnancy by

(1) Ovary and Pituitary gland

(2) Secondary oocyte and Graafian follicle

(3) Placenta and Pars nervosa

(4) Corpus luteum and Placenta

170. Lippes loop and LNG-20 respectively are

(1) Hormone releasing IUD and Cu^{++} releasing IUD

(2) Non-medicated IUD and hormone releasing IUD

(3) Non-medicated IUD and oral pill

(4) Implant and hormone releasing IUD

171. All of the following diseases are not completely curable even if detected early and treated properly, **except**

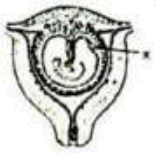
(1) HIV infection

(2) Genital Herpes

(3) Chlamydiasis

(4) Hepatitis-B

172. Identify the structure marked as 'X', and select the correct option



(1) Embryo

(2) Yolk sac

(3) Umbilical cord

(4) Placental villi

173. Consider the given statements and select the correct option.

Statement A : The level of FSH increases several fold in the maternal blood during pregnancy.

Statement B : The blastomeres in the blastocyst are arranged into an outer layer called inner cell mass.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

174. After implantation, finger like projections appear on the trophoblast called _____ which are surrounded by the uterine tissue and maternal blood.

Select the **correct** option that fills the blank.

(1) Umbilical Cord

(2) Uterine issue

(3) Yolk sac

(4) Chorionic villi

175. Match following columns A and B.

	Column A		Column B
a.	Human heart is formed	(i)	During 5th month of gestation
b.	Development of limbs and digits	(ii)	At the end of first trimester of pregnancy
c.	Major organ systems are formed	(iii)	By the end of 2nd month of pregnancy
d.	First movement of foetus is observed	(iv)	After one month of pregnancy

Choose the option containing all **correct** match.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

176. Number of ova produced from 10 primary oocytes is _____ while number of spermatids produced from 10 secondary spermatocytes is _____.

Select the correct option that fills the blanks respectively.

(1) 10, 10

(2) 20, 10

(3) 10, 20

(4) 20, 20

177. Which of the following ARTs involves direct injection of sperm into the ovum?

(1) IUI

(2) GIFT

(3) ZIFT

(4) ICSI

178. Select the **odd** one w.r.t. the acellular layer present in Graafian follicle,
(1) It is known as zona pellucida
(2) Thick coat composed of glycoprotein
(3) Synthesized by secondary oocyte
(4) Causes disintegration of the endometrium leading to menstruation

179. All of the following are considered as methods to prevent spread of venereal diseases, **except**
(1) Avoiding sex with unknown partner(s)
(2) Always using condom during coitus
(3) Consulting a doctor for early detection
(4) Using cervical caps during coitus

180. Which of the following hormones is involved in the process of child birth?
(1) Cortisol from adrenal medulla
(2) Prolactin from neurohypophysis
(3) Thyroxine from thymus
(4) Oxytocin from pars nervosa

181. Oogenesis in females begins
(1) During puberty
(2) After puberty
(3) Before birth
(4) After birth

182. Which of the following does **not** occur by the end of 2nd trimester?
(1) The body is covered by fine hairs
(2) Eyelashes are formed
(3) Eyelids separate
(4) Foetus is ready for delivery

183. In human females, which of the following layers of uterine wall undergo cyclic changes during menstrual cycle?
(1) Endometrium
(2) Myometrium
(3) Perimetrium
(4) Myometrium and perimetrium

184. Which of the following is correctly related to lecithin?
(1) It is a phosphorylated glyceride found in cell membrane
(2) Its structure contains three fatty acids
(3) Two nitrogen atoms are attached to phosphoric acid in its structure
(4) It is an esterified fatty acid found in nucleotides

185. Select the correct match w.r.t. metabolites which do not play role in normal physiological processes in host organisms but many of them are useful to 'human welfare'.
(1) Alkaloids → Morphine, Ricin
(2) Terpenoides → Monoterpenes, Concanavalin A
(3) Drugs → Vinblastin, Curcumin
(4) Pigments → Carotenoids, Codeine

Aakash + BYJU'S

SECTION-B

186. The opinion of how many registered medical practitioners is required for MTP, if the pregnancy has lasted for 15 weeks?
(1) 1
(2) 2
(3) 4
(4) 3

187. All the following options are correct w.r.t. implants, **except**
(1) They inhibit ovulation and implantation
(2) They alter the quality of cervical mucus
(3) They are placed under the skin
(4) They contain combinations of non-steroidal hormones

188. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Lactational amenorrhea	(i)	Copper releasing IUD
b.	Vaults	(ii)	Progestogen-estrogen combination
c.	Multiload-375	(iii)	Natural method of contraception
d.	Emergency contraceptive	(iv)	Barriers made up of rubber

(1) a(ii), b(i), c(iv), d(iii)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(ii), c(i), d(iii)
(4) a(iii), b(iv), c(i), d(ii)

189. Read the statements given below and choose the **correct** one.
(1) Permissible use of the technique amniocentesis in India is for detecting sex of the unborn foetus.
(2) Artificial insemination implies transfer of sperms of a healthy donor to a test tube containing many ova.
(3) Use of 'Nirodh' prevents sperms from reaching cervix.
(4) Generally chances of conception are nil as long as mother breast feeds the young one.

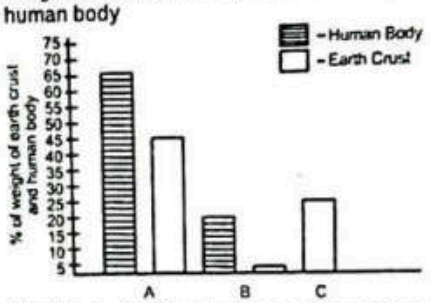
190. How many of the following are methods of contraception for males?
a. Condoms
b. Diaphragm
c. Lactational amenorrhea
d. Implants
e. Vasectomy
Select the **correct** option
(1) 2
(2) 3
(3) 1
(4) 5

191. Select the hormones that are produced by placenta in females during pregnancy and choose the **correct** option.
(1) Only hCG
(2) Only progesterone
(3) Only estrogen
(4) Both hCG and hPL

192. A cushion of fatty tissue covered by skin and pubic hair in female external genitalia is known as
(1) Mons pubis
(2) Hymen
(3) Labia minora
(4) Clitoris

193. A diglyceride refers to
(1) Dihydroxy propane
(2) Dihydroxy ethane
(3) Glycerol esterified with two fatty acids at two different carbons
(4) Glycerol molecule esterified with two phosphate moieties

194. Analyze the graph given below w.r.t the % of inorganic elements in the earth's crust and human body



Identify A, B, C and choose the **correct** statement.
(1) A represent the element Oxygen (O)
(2) B represent the element Silicon (Si)
(3) C represent the element Carbon (C)
(4) B represent the element Nitrogen (N)

195. The function of 'A' epithelium is to move particles or mucus in a specific direction and is mainly present in the inner surface of hollow organs like 'B'?
Select the correct option for 'A' and 'B'.

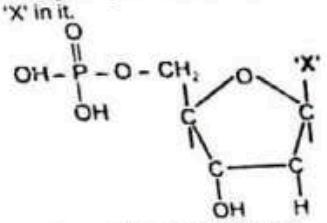
	A	B
(1)	Cuboidal	PCT of nephron
(2)	Ciliated	Bronchioles
(3)	Cuboidal	Stomach
(4)	Columnar	Intestine

(1) (1)
(2) (2)
(3) (3)
(4) (4)

196. How many of the following statements are correct w.r.t. Biomolecules?
a. Arachidonic acid has 19 carbon atoms excluding the carboxyl carbon.
b. Gingelly oil has lower melting point than fat.
c. Paper made from plant pulp and cotton fibres is cellulosic.
d. The acid insoluble pool represents roughly the cytoplasmic composition.
Select the **correct** option.
(1) Three
(2) Four
(3) One
(4) Two

197. The epithelium which has limited role in secretion and absorption is present in how many of the given locations?
(a) Dry surface of skin
(b) Moist surface of buccal cavity
(c) Inner lining of ducts of pancreas
(d) Air sacs of lungs
(e) Tubular parts of nephron
Select the **correct** option.
(1) Three
(2) Four
(3) Five
(4) Two

198. Identify the category of compound shown in the figure given below and the component 'X' in it.



Choose the **correct** option.

	Category		Component 'X'
(1)	Nucleotide	–	Uracil
(2)	Nucleoside	–	Thymine
(3)	Nucleotide	–	Thymine
(4)	Nucleoside	–	Uracil

(1) (1)
(2) (2)
(3) (3)
(4) (4)

199. Select the statement which is **not** related to smooth muscle fibres.
(1) Their fibres taper at both ends and do not show striations.
(2) They are located in the wall of blood vessels, stomach and intestine.
(3) Their functions cannot be directly controlled by our will.
(4) They show multinucleated appearance and branching.

200. Amino acid whose 'R' group is basic in nature is
(1) Lysine
(2) Valine
(3) Glycine
(4) Serine

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